DBMS -- Keys & Constraints

Target: Explain PK vs UK vs FK in 40s. Know the full key hierarchy cold.

The Key Hierarchy

```
Super Key  (broadest -- any combo that gives uniqueness, can be redundant)
      Candidate Key  (minimal super key -- no column can be removed)
              Primary Key    (chosen candidate key; NOT NULL, one per table)
              Alternate Key  (remaining candidate keys not chosen)
              Unique Key     (like PK but allows one NULL; multiple per table)
Foreign Key     links to another table's PK (referential integrity)
Composite Key   two or more columns together form the key
```

One Table, All Keys Explained

Table: students

StudentID	Email	Phone	Name
101	a@gmail.com	9999	Avanish
102	b@gmail.com	8888	Rahul

Key Type	Example	Why
Super Key	{StudentID}, {Email}, {StudentID, Email}	All uniquely identify a row
Candidate Key	{StudentID}, {Email}, {Phone}	Minimal -- can't drop any column
Primary Key	{StudentID}	Chosen; no NULL allowed
Alternate Key	{Email}, {Phone}	Candidate keys not chosen
Unique Key	{Email}	Enforces uniqueness, allows one NULL

PK vs UK vs FK -- Your 40-Second Script

"PK uniquely identifies every row -- no NULLs allowed, and there can only be one per table. UK also enforces uniqueness but allows one NULL, and you can have multiple UKs in a table. FK is a column in one table that points to the PK of another table -- it enforces referential integrity, meaning you can't reference a row that doesn't exist."

Primary Key vs Unique Key -- Side by Side

Feature	Primary Key	Unique Key
Uniqueness	[OK]	[OK]
NULL allowed	Never	[OK] One NULL per column
Count per table	Only one	Multiple allowed
Index created	Clustered index	Non-clustered index
Can be FK target	[OK] Yes	[OK] Yes (in most DBs)

Foreign Key -- Referential Integrity

```sql
-- Parent table
CREATE TABLE departments (
  dept_id INT PRIMARY KEY,
  dept_name VARCHAR(50)
);
-- Child table -- dept_id references parent
CREATE TABLE employees (
  emp_id   INT PRIMARY KEY,
  name     VARCHAR(50),
  dept_id  INT,
  FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);
-- This INSERT will FAIL -- dept_id 999 doesn't exist in departments
INSERT INTO employees VALUES (1, 'Avanish', 999);
```

FK behaviors on DELETE/UPDATE of parent row:

Action	What happens to child rows

CASCADE
Child rows deleted/updated automatically

SET NULL
Child FK column set to NULL

RESTRICT
Delete blocked if child rows exist

NO ACTION
Same as RESTRICT (default in most DBs)

Composite Key (also called Compound Key)

Two or more columns that together form a unique key
Neither column alone is unique
The ebook calls this a Compound Key -- same concept, different name used in different books

sql
```
-- Neither student_id nor course_id alone is unique
-- But (student_id, course_id) together is unique per enrollment
CREATE TABLE enrollments (
  student_id INT,
  course_id  INT,
  enrolled_on DATE,
  PRIMARY KEY (student_id, course_id)   -- composite/compound PK
);
```

Surrogate Key

An artificial key added purely to uniquely identify a record
Not derived from any real-world/business data
Usually an auto-incremented integer or UUID

Why use it?
- Natural keys (PAN, Aadhaar) can change or be long -- bad for joins
- Surrogate keys are stable, short, numeric -- ideal as PK

sql
```
CREATE TABLE employees (
  emp_id   INT AUTO_INCREMENT PRIMARY KEY,  -- surrogate key
  pan_no   VARCHAR(10) UNIQUE,              -- natural key (business meaning)
  name     VARCHAR(50)
);
```
Natural key vs Surrogate key:

Natural Key
Surrogate Key

Source
Real-world data (PAN, email)
Artificially generated (1, 2, 3...)

Meaningful?
Yes
No (just an ID)

Can change?
Yes (risky)
No (stable)

Example
Aadhaar number
auto-increment id

The 6 Constraints -- Quick Reference

Constraint	Rule	Example
NOT NULL	Column must always have a value	name VARCHAR(50) NOT NULL
UNIQUE	No duplicate values in column	email VARCHAR(100) UNIQUE
PRIMARY KEY	NOT NULL + UNIQUE; one per table	id INT PRIMARY KEY
FOREIGN KEY	Value must exist in referenced table	FOREIGN KEY (dept_id) REFERENCES dept(id)
CHECK	Custom condition must be true	CHECK (age >= 18)
DEFAULT	Auto-fills value when none provided	status VARCHAR(10) DEFAULT 'active'

Memory trick: N-U-P-F-C-D   "Never Use Primary Foreign Checks Defaultly"

Primary Key -- SQL Examples (from ebook)
CREATE TABLE with PK
sql
```
CREATE TABLE PERSONS (
  ID        INT NOT NULL,
  LastName  VARCHAR(255),
  FirstName VARCHAR(255),
  Age       INT,
  PRIMARY KEY (ID)
);
```
Add PK using ALTER TABLE
sql
```
ALTER TABLE table_name
ADD CONSTRAINT constraint_name
PRIMARY KEY (column1, column2, ...);
```

Foreign Key -- SQL Examples (from ebook)
FK on CREATE TABLE
sql

```
CREATE TABLE ORDERS (
  order_ID     INT NOT NULL,
  orderNumber  INT NOT NULL,
  personID     INT,
  PRIMARY KEY (orderID),
  FOREIGN KEY (personID) REFERENCES persons(personID)
);
```

Rule: FK column (personID) and PK it references (persons.personID) must have the same data type.

All Key Types -- Final Summary

Key
Definition
NULL?
Count per table

Super Key
Any combo that gives uniqueness
Allowed
Many

Candidate Key
Minimal super key
Not recommended
Multiple

Primary Key
Chosen candidate key
Never
One only

Alternate Key
Non-chosen candidate keys
Allowed
Multiple

Unique Key
Enforces uniqueness
[OK] One NULL
Multiple

Foreign Key
References PK of another table
[OK] Allowed
Multiple

Composite Key
2+ columns forming a key together
Depends
Multiple

Surrogate Key
Artificial auto-generated key
Never
One (usually)

Follow-Up Questions (Expect These)
Q: Can a table have no primary key?

Yes, technically. But it's bad practice -- duplicates become possible and joins become unreliable.

Q: Can a FK reference a Unique Key instead of a PK?

Yes, in most RDBMS (PostgreSQL, SQL Server). The referenced column just needs to be unique.

Q: What is a surrogate key?

An artificial key added purely for identification (e.g., auto-increment id). Not derived from business data. Preferred when natural keys are long or composite.

Q: What is a natural key?

A key that comes from real-world data (e.g., Aadhaar number, PAN). Risky if the data can change.